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Design and synthesis of novel 8-(azaindolyl)-benzoazepinones as potent and selective ROCK inhibitors†

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We report the characterization of potent and selective ROCK inhibitors identified through a core-hopping strategy. A virtual screening workflow, combining ligand- and structure-based methods, was applied on a known series of ROCK inhibitors bearing an acetamido-thiazole scaffold. The most promising replacement of the central core was represented by a benzoazepinone ring, which was selected as a starting point for a subsequent chemical exploration. The overall design approach exploited previous SARs available for congeneric series and crystallographic information to optimize the hinge-binding group as well as the terminal aromatic moiety interacting with the glycine-rich loop. The introduction of elongated and flexible charged groups led to compound **15**, which exhibited sub-nanomolar potencies in biochemical and cellular assays, as well as a remarkable selectivity over PKA. HDX studies not only supported the postulated binding mode of compound **15** but also confirmed the crucial role of specific ROCK peptide segments in driving ligand selectivity.

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Introduction

Rho-associated protein kinases (ROCKs) are ubiquitous serine–threonine protein kinases which regulate a variety of cellular functions. The ROCK kinase family comprises two highly-homologous isoforms, namely ROCK1 and ROCK2, which exhibit more than 90% sequence identity in the kinase domain.[1] The association of ROCK with the small GTP-binding protein RhoA promotes the phosphorylation of multiple substrates and the subsequent activation of different signalling pathways.[2] In particular, ROCK-mediated phosphorylation of myosin light chain phosphatase (MLCP) and myosin light chain (MLC) triggers actin cytoskeletal changes and regulates the shape and function of the actomyosin complex. Given its crucial role in smooth muscle contraction as well as in actin dynamics and reorganization, dysregulation of the RhoA/ROCK downstream signalling has been associated with a variety of pathological conditions, including asthma,[3] pulmonary hypertension,[4] glaucoma[5] and cardiovascular diseases.[6] Moreover, a growing body of evidence has shown additional roles of the Rho/ROCK pathway in other pathological processes, such as cancer,[7] osteoporosis,[8] neuronal degeneration[9] and Alzheimer's disease.[10] For this reason, small-molecule compounds able to inhibit ROCK activity represent promising therapeutic agents for the treatment of a wide range of disorders.

Medicinal chemistry research has led to the development of several classes of ROCK inhibitor,[11] some of which have successfully reached the market or have entered clinical trials. Fasudil was approved for cerebral vasospasm[12] and it has been clinically investigated for the treatment of amyotrophic lateral sclerosis[13] and Raynaud's phenomenon,[14] whereas ripasudil[15] and netarsudil[16] have been marketed for glaucoma and ocular hypertension. Belumosudil, previously known as SLx-2119/KD-025, is a ROCK2-selective inhibitor currently undergoing clinical trials for the treatment of several conditions, including idiopathic pulmonary fibrosis,[17] psoriasis[18] and graft-*versus*-host disease.[19] However, despite being widely applied in many different pathological conditions, ROCK inhibitors often suffer from apparent limitations, including sub-optimal potencies or poor selectivity profiles, which might be responsible for known off-target and toxic effects.[20–23] Thus, despite the long-standing interest in ROCK inhibitors, the pursuit of novel small-molecule agents exhibiting high potency and selectivity still represents an attractive subject of research.

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Results and discussion

Recently, many research groups have successfully exploited the increasing amount of structural information available for ROCK as well as for different closely related AGC-family kinases to design potent and selective compounds. In particular, several ROCK ligands devoid of activity towards PKA, a key off-target of non-specific ROCK inhibitors, have been developed by leveraging structural differences between ATP-binding cavities.[24–27] For example, a series of 3-substituted 7-azaindole derivatives, exhibiting ROCK1 K_i values in the nanomolar range, as well as notable selectivity over PKA, has been recently reported by Vertex.[28] The most representative compound (compound **1**, Fig. 1) exhibited a ROCK1 pK_i higher than 9 and a 570-fold selectivity over PKA. In this study, a two-step design strategy was applied to optimize a promising series of inhibitors bearing an acetamido-thiazole scaffold. Specifically, the hinge-binding motif was first rigidified through a cyclization of an amino-pyrimidine scaffold. Subsequently, a non-conserved kinase segment in ROCK and PKA was targeted using elongated solubilizing groups branching from the central phenyl ring, leading to a dramatic increase in selectivity.

We initially focused on the X-ray structure of human ROCK1 co-crystallized with the 3-substituted 7-azaindole derivative **2** (Fig. 1, center) developed by Vertex (PDB ID: 5KKS).[28] A visual inspection of the ATP-binding cavity clearly highlighted peculiar ligand features. The azaindole moiety formed a bidentate interaction with the backbone carbonyl of Glu154 and the backbone NH group of Met156 (see Fig. S4†). Moreover, the acetamido-thiazole linker not only formed an extensive network of interactions with neighbouring residues

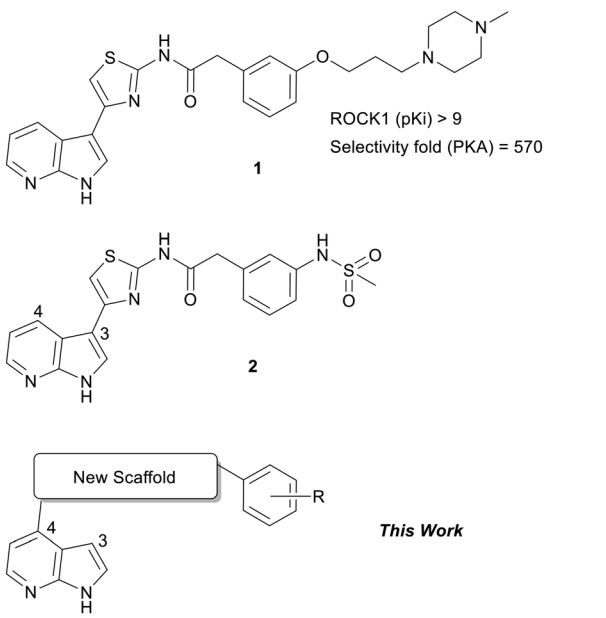


Fig. 1 Chemical structures of representative Vertex compounds **1** and **2**. Schematic depiction of the medicinal chemistry strategy pursued in this work (bottom).

but also pre-organized the ligand conformation, forcing the terminal phenyl ring into an "out-of-plane" geometry which was well accommodated in a small volume beneath the glycine-rich loop. With the aim at identifying novel cores linking the 4-position of the azaindole hinge-binder to the terminal *N*-phenylmethanesulfonamide group (branches), we devised a novel virtual screening protocol which combined ligand- and structure-based information (Fig. 2).

Briefly, the 2-acetylamino-thiazole moiety was deleted from the three-dimensional (3D) structure of the co-crystallized compound and the disconnected branches were then joined using a customized database of linkers. Specifically, a large compound collection available in the Schrodinger Suite 2020. Compound **2** was first processed with the corefinder utility to identify and extract potential core structures. Subsequently, all identified cores having two or more disconnections were attached to the 4-position of the azaindole ring and to the original attachment position on the *N*-phenylmethanesulfonamide fragment. During this phase, all possible linking combinations, generated by joining each disconnection on the core scaffold to each branch, were included to maximize the chemical exploration. The resulting 131 266 compounds were then evaluated by their ability to mimic the binding orientation of compound **2**. To this end, a shape screening campaign was conducted with Phase,[29,30] using the crystallographic pose of compound **2** as reference. To further evaluate the quality of the alignment, the root mean square deviation (RMSD) value of the atoms composing the branch moieties in each aligned compound was calculated with respect to the reference co-crystallized inhibitor. Only those compounds showing an RMSD value equal to or lower than 2 Å were subsequently subjected to a docking procedure, where ligands were energy-minimized in the confines of the active site and scored using default Glide settings.[31,32] The 20 best-scored docking solutions according to the docking score function were submitted to a binding free-energy evaluation applying the MM-GB/SA approach implemented in Prime.[33,34]

Interestingly, the best-ranked derivative contained a unexpected benzoazepinone moiety (compound **3** was the best-scored compound identified by this campaign), which formed extensive hydrophobic interactions with different amino acids delimiting the ATP-binding cavity, including

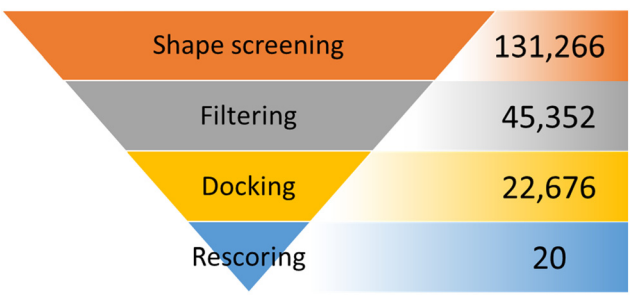


Fig. 2 Schematic representation of the virtual screening workflow applied in this study.

Gly83, Val90 and Leu205 (Fig. 3). Moreover, the carbonyl oxygen of the azepinone ring superposed well with the amide oxygen of the co-crystallized compound **2**, allowing the formation of a productive hydrogen-bond (H-bond) interaction with the catalytic lysine 105. Intriguingly, a small series of partially saturated heterocycles has been recently evaluated as alternative core structures to identify novel ROCK inhibitors.[35]

The authors identified a benzooxazepinone-containing derivative as the most promising compound, further supporting the application of these unique fused systems to selectively targeting ROCK kinases.

Thus, given its novel and favourable structural features, a first round of compounds bearing a benzoazepinone core was prepared to assess the ability of this substructure to mimic the original amido-thiazole linker (Table 1).

Inhibitory potencies of synthesized compounds against ROCK1 and ROCK2 were evaluated in two biochemical assays employing different ATP concentrations. The standard assay employed a concentration of ATP ([ATP]) close to K_m, providing potency values which were used to infer preliminary SAR, and to readily identify the most promising analogues. A customized protocol was conducted with [ATP] $\gg K_m$, which was specifically conceived to mimic the high intracellular ATP concentrations *in vivo*, permitting the accurate evaluation of potent compounds that might have reached the activity plateau in the standard protocol.[36]

According to our experience with ROCK assays, standard and customized settings are often in good agreement and usually interchangeable during SAR evaluations when data is expressed as K_i. The selectivity of ROCK inhibitors over PKA has been used as a surrogate indicator of their ability to achieve specificity against a broader panel of kinase targets,[37] hence an additional biochemical assay has been implemented for protein kinase A (PKA).

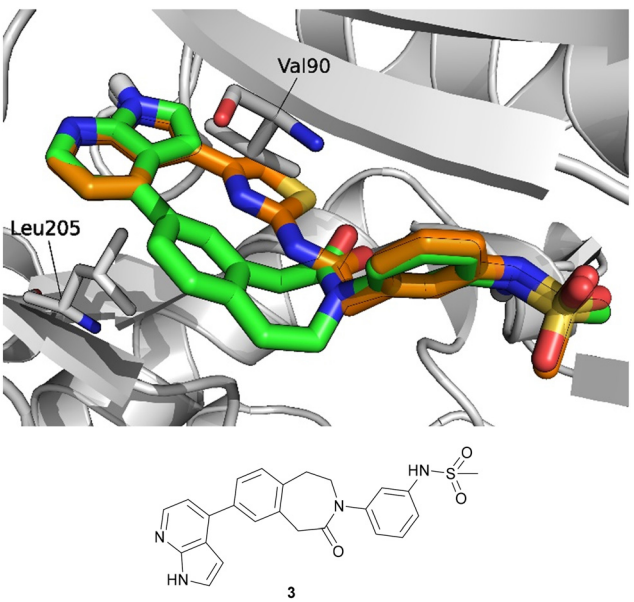


Fig. 3 Superposition of the MM-GB/SA-optimized solution of compound **3** (green carbons) and the co-crystallized pose of compound **2** (transparent orange carbons – PDB: 5KKS) and structure of compound **3**.

We initially synthesized and tested compound **3**, which represented the proof-of-concept design of our core-hopping strategy. This derivative exhibited sub-nanomolar potency in both ROCK1 and ROCK2 biochemical assays, validating the benzoazepinone core as a suitable replacement of the Vertex linker moiety.

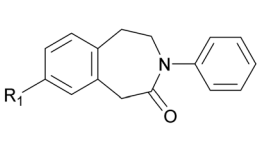
Given the ability of the new scaffold to retain a pronounced ROCK inhibitory activity, we then sought to thoroughly investigate the effect of different terminal branches on potency and selectivity. As a first step, the terminal sulfonamide group was removed, yielding the parent compound **4**. This derivative was specifically designed to include only the essential pharmacophoric features identified in the Vertex series, *i.e.*, an azaindole hinge-binder and a phenyl ring able to be accommodated beneath the glycine-rich loop. Compound **4** showed only a slight loss in potency compared to derivative **3**, indicating a negligible influence of the terminal sulfonamide fragment on compound activity. Accordingly, the ligand efficiency (LE) metric calculated for compound **4** was slightly higher compared to that of compound **3**, further confirming the ability of a minimally functionalized benzoazepinone core to recapitulate the essential pharmacophore of ROCK inhibitors.

That said, compound **4** exhibited a significant PKA inhibitory effect, resulting in a negligible selectivity for ROCK over PKA. Crystallographic studies confirmed the postulated binding mode of this novel series and also provided valuable insights for the design of novel inhibitors with improved pharmacological and physicochemical profiles. Thus, a co-crystal structure of inhibitor **4** with human ROCK2 was obtained (Fig. 4).

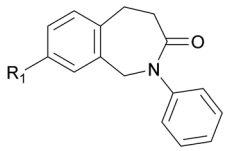
Notably, the X-ray pose of compound **4** closely resembled the docking solution of compound **3** (Fig. 3), with the azaindole moiety being involved in a bidentate H-bond interaction with the hinge region and the terminal phenyl ring being accommodated in a small, lipophilic cavity beneath the glycine-rich loop. This indicated, once again, that the distinctive U-shaped geometry of the benzoazepinone ring can present the pendant groups in a binding-competent conformation, independently from their substitution patterns. Remarkably, the central core scaffold was rotated about 40 degrees counterclockwise with respect to the azaindole hinge-binder, leading to a minimum-energy geometry (Fig. 4) able to retain the key H-bond interaction with the catalytic lysine. To further confirm the ability of this specific core moiety to correctly orient the crucial ligand pharmacophoric features within the ROCK active site, its regioisomer **5** was synthesized and tested. As expected, the striking change in the relative orientation of ligand functional groups led to a dramatic decrease in potency compared to the original substitution pattern.

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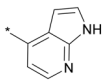
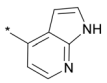
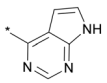
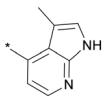
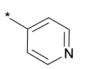
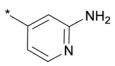
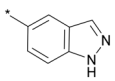
Table 1 Initial SAR exploration around the hinge-binding group



Scaffold **A**



Scaffold **B**

Compound (scaffold)	R_1	ROCK1 (K_i[a]) / ROCK2 (K_i[a]) / PKA (K_i[a])	Fold sel. PKA Over ROCK 1 & 2	LE[b]
3	—	0.15 ± 0.01[d] 0.15 ± 0.03[d] nd	nd nd	0.37[c]
4 (A)		0.42 ± 0.02[d]/2.0 ± 0.1[c] 0.2 ± 0.1[d]/1.0 ± 0.4[c] 0.5 ± 0.1	1.3[d] 2.5[d]	0.42[d] 0.45[c]
5 (B)		126 ± 26[c] 159 ± 31[c] nd	nd nd	0.33[c]
6 (A)		7.9 ± 0.8[d] 2.5 ± 0.1[d] 5 ± 2	0.6[d] 0.2[d]	0.37[d]
7 (A)		10 ± 2[c] 8 ± 2 nd	nd nd	0.38[c]
8 (A)		251[d] 316[d] 200 ± 2	0.8[d] 0.6[d]	0.29[d]
9 (A)		50 ± 12[d] 16 ± 8[d] 32 ± 3	0.6[d] 0.2[d]	0.36[d]
10 (A)		79 ± 9[d] 25 ± 5[d] 794 ± 248	10[d] 32[d]	0.37[d]
11 (A)		251[d] 316[d] 501 ± 133	2.0[d] 1.6[d]	0.26[d]

[a] Values reported as nM. [b] Ligand efficiency was calculated as (1.4 × pK_i ROCK2)/heavy atom count. [c] K_i values obtained with [ATP] = K_m. [d] K_i values obtained with [ATP] ≫ K_m; nd: not determined.

With the aim at further exploring SARs around the hinge-binding group, different derivatives were prepared by replacing or decorating the azaindole system. The insertion of an additional nitrogen atom on the azaindole ring (compound **6**) brought a limited decrease in both ROCK and PKA potencies, without affecting the overall selectivity ratio. Similarly, the introduction of a methyl group at position 3 of the azaindole hinge binder (compound **7**) did not cause a significant drop in cell-free inhibitory effect. This could be rationalized by considering the torsional energy profile of a surrogate compound bearing the same methyl substituent (Fig. 5). Indeed, even though the introduction of the methyl functionality led to a slight shift of the energy minima as well as a significant enhancement of energy barriers corresponding to planar configurations compared to the unsubstituted derivative, the dihedral angle observed in the co-crystallized ligand pose was still located in a minimum-energy region, indicating that the substituted compound can still assume a binding-competent geometry with a negligible energy penalty.

The deletion of the pyrrole ring of the azaindole hinge-binder (compound **8**) removed one of the key H-bond functionalities involved in the bidentate interaction with the hinge region of ROCK, causing a dramatic loss in potency in cell-free assays. With the aim at restoring the original bidentate interaction, a 2-amino functionality was installed on the hinge-binding group of compound **8**, leading to derivative **9**. As hypothesized, the recovery of the bidentate H-bond brought an increase in inhibitory effect compared with the monodentate hinge binder of compound **8**, although not comparable with the original azaindole-containing inhibitor **4**. Given the importance of the bidentate H-bond interaction with the hinge region, the azaindole was replaced with a pyrazole (compound **10**), a monocyclic ring system

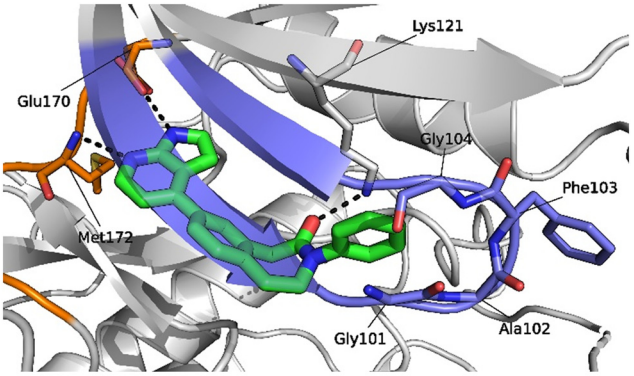


Fig. 4 Co-crystal structure of compound **4** bound to human ROCK2. (2.63 Å, PDB ID: 9EP8) and structure of compound **4**.

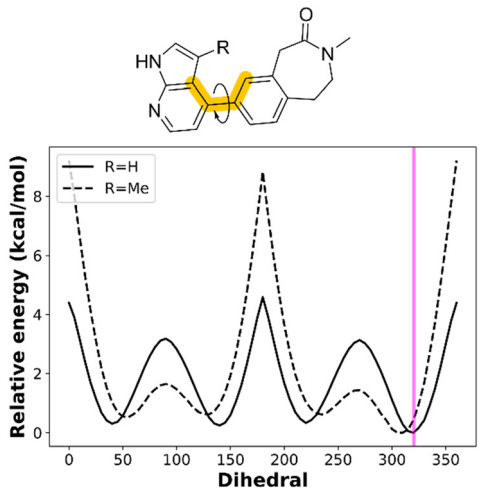


Fig. 5 Potential energy profiles in the gas phase along the highlighted dihedral angle for two model compounds at the B3LYP-D3/6-31G** level. The torsion angle of the co-crystallized pose is reported as a transparent magenta line.

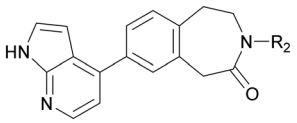
containing both a H-bond acceptor and donor functionality, which has been incorporated into many different classes of ROCK inhibitors.[11] Interestingly, the marked decrease in potency observed for compound **10** was more pronounced for PKA than ROCK, leading to an increase in selectivity. That said, pyrazole compound **10** showed a LE similar to aminopyridine compound **9**, indicating that the loss in inhibitory activity was partially compensated by the incorporation of H-bond donor and acceptor features within a smaller ring system. The limited effect on selectivity observed for compound **10** prompted us to focus on indazole, another pyrazole-containing hinge binder which has been commonly applied in the design of numerous ROCK inhibitors.[11] However, the indazole-based inhibitor compound **11** proved to be one of the least active and selective compounds of this subset, exhibiting biochemical potencies in the high-nanomolar range for both ROCK isoforms. The decrease in potency observed for compound **11** could be rationalized by considering that the additional phenyl spacer between the pyrazole ring and the benzoazepinone core might cause a shift of the central scaffold as well as of the terminal phenyl ring towards the outer portion of the ATP-binding cavity, weakening crucial interactions with the catalytic lysine and with the tip of the glycine-rich loop.

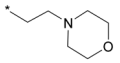
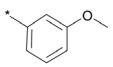
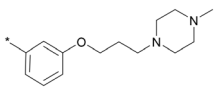
In conclusion, the unsubstituted azaindole ring yielded the most potent inhibitor and was thus retained throughout the subsequent chemical exploration.

The second optimization step was focused on substituents branching from the benzoazepinone nitrogen (Table 2). Given the fundamental role of the terminal phenyl ring of inhibitor **4** in stabilizing the glycine-rich loop of ROCK, it was not surprising that its deletion (compound **12**) resulted in a decrease in inhibitory potency for both ROCK isoforms, without any notable effects on the overall selectivity profile. To restore the interaction with the glycine-rich loop, we introduced an elongated *N*-ethyl-morpholine group on the benzoazepinone ring (compound **13**). However, this substituent did not improve the overall biochemical profile, probably due to its high flexibility and sp^3-rich character, which might hamper the formation of stable interactions underneath the glycine-rich segment, and/or to its weak basicity (calculated pK_a: 6.7), that might lead to energetic penalties caused by the accommodation of the terminal hydrophilic ring under the hydrophobic tip of the same protein loop. Given the sub-optimal results with alkyl chains, the unsubstituted phenyl ring of inhibitor compound **4** was considered a more attractive starting point for further functionalization. The co-crystal structure of compound **4** bound to human ROCK2 clearly pinpointed the *meta* position as the preferred exit vector for the introduction of additional substituents, as *para* and *ortho* substituents were predicted to clash with neighbouring protein residues. As a first step, we evaluated the effect produced by the introduction of a simple methoxy group, a chemical handle which could also grant access to a wide range of synthetically accessible methyl group replacements and that has already been introduced in several series of ROCK inhibitors.[28,38] Notably, the *m*-methoxyphenyl derivative compound **14** exhibited single-digit nanomolar potencies against both ROCK isoforms, further confirming the ability of *meta* substituents to enhance inhibitory activity. As already mentioned, recent medicinal chemistry campaigns aimed at identifying novel ROCK inhibitors have successfully exploited a non-conserved kinase region, located on the αC-helix, to gain significant selectivity over PKA.[28] In particular, the aromatic moiety accommodated beneath the glycine-rich loop was further functionalized with flexible and positively-charged groups.[28,39] The introduction of a terminal basic centre not only allowed the formation of a strong ionic interaction with

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Table 2 SAR exploration around the benzoazepinone substituents



Compound	R_2	ROCK1 (K_i[a]) / ROCK2 (K_i[a]) / PKA (K_i[a])	Fold sel. PKA Over ROCK 1&2	LE[b]
12	Me	25 ± 8[c] / 5 ± 2[c] / 40 ± 10	1.6[c] / 7.9[c]	0.42[c]
13		20 ± 1[c] / 5.0 ± 0.4[c] / 7.9 ± 0.3	0.4[c] / 1.6[c]	0.33[c]
14		1.60 ± 0.01[c] / 1.3 ± 0.6[c] / 5.0 ± 0.3	3.2[c] / 4.0[c]	0.36[c]
15		0.28 ± 0.02[c] / 0.13 ± 0.02[c] / 20 ± 3	63[c] / 159[c]	0.31[c]

[a] Values reported as nM. [b] Ligand efficiency was calculated as (1.4 × pK_i ROCK2)/heavy atom count. [c] K_i values obtained with [ATP] ≫ K_m; nd: not determined.

a ROCK-specific aspartate residue (Asp117 ROCK1, Asp133 ROCK2), which is absent in PKA, but it also may converge with chemical strategies aimed at increasing compound solubility. The inclusion of an elongated piperazine tail on the terminal phenyl ring of our series led to inhibitor compound **15**, which was characterized by sub-nanomolar biochemical potencies at both ROCK isoforms. Notably, this compound showed the best selectivity profile of the series, exhibiting 63/159-fold selectivity for ROCK1 and 2 isoforms over PKA, respectively.

To elucidate the molecular determinants responsible for the selectivity profile of compound **15** and to validate our initial design hypothesis on the engagement of the αC-helix, we conducted HDX-MS experiments to compare the deuterium uptake between the ROCK2-compound **15** bound state and ROCK2 unbound state (Fig. 6). The ROCK2-compound **4** state was also included as a reference system to better dissect the effect of the elongated solubilizing tail on protein dynamics. As illustrated in the difference chart both compounds **4** and **15** gave a significant decrease in deuteration for amino acids belonging to the glycine-rich loop (highlighted in red, see also Fig. S2-a†) and to the αC-helix (highlighted in green, see also Fig. S2-b†), whereas no effects on deuterium uptake were detected for peptides residing in the C-tail and in the activation loop of ROCK2 (coloured in gray and pink, respectively, see also Fig. S2-c and -d† respectively). Interestingly, the decrease in deuteration observed for the glycine-rich loop and for the αC-helix was more pronounced for inhibitor **15** than its shorter analogue **4**. The different stabilization effect exhibited by compounds **15** and **4** could be also detected by analysing the time-evolution of deuterium uptake for overlapping peptides belonging to the glycine-rich segment and to the αC-helix (Fig. S2a–d and S3,† respectively).

Again, these plots clearly showed that the decreased deuteration induced by compound **15** was consistently higher than that caused by **4**, especially at longer time points. Taken together, these results clearly demonstrated the involvement of the glycine-rich segment as well as of the αC-helix of ROCK2 in the stabilization of inhibitor **15**, supporting our initial design hypothesis. Moreover, this study further strengthened the growing body of evidence pinpointing the αC-helix and, specifically, Asp117/Asp133 as a crucial ROCK feature involved in selectivity. Intriguingly, both derivatives **4** and **15** caused a remarkable increase in deuterium uptake in the so-called p+1 loop[40] (residues 253–262), a highly conserved kinase segment known to play a key role in substrate recognition.[41] Consequently, the ligand-induced increase in p+1 loop flexibility might thus hamper the stabilisation of the natural peptide substrate, ultimately leading to the significant inhibitory effect observed for both compounds.

As a next step, a subset of the most interesting derivatives was tested in a cell-based assay aimed at evaluating the compounds' ability to prevent MLC phosphorylation in a PASM (pulmonary artery smooth muscle) cell line (Table 3).

Selected compounds showed cell-based potencies ranging from low-nanomolar to sub-micromolar values, which was consistent with their corresponding inhibitory activities in cell-free assays.

Compounds **4** and **15** showed the highest cell potencies and were thus progressed to a preliminary ADME panel including kinetic solubility, permeability in Caco-2 monolayers and metabolic stability in rat microsomes

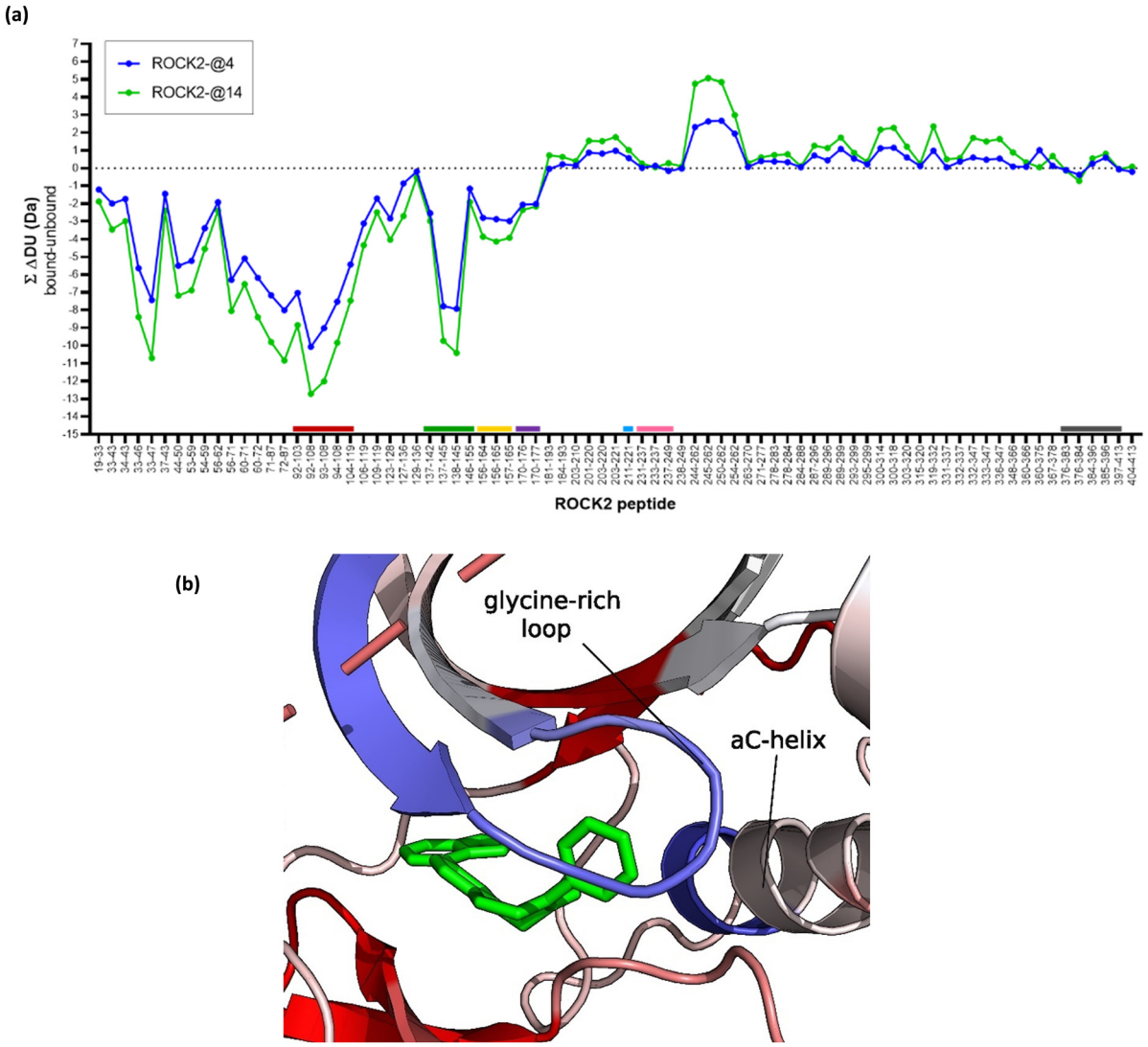


Fig. 6 (a) Difference plot illustrating the difference in hydrogen/deuterium exchange between the bound and unbound state of ROCK2. The summed differences in deuterium uptake registered over all six HDX timepoints have been plotted for each peptide ($\sum\Delta$DU at 10 s, 30 s, 5 min, 10 min, 30 min, 1 h). Negative difference values indicate that a decrease in DU occurs in the bound state (protection *versus* exchange), whereas positive values are a sign of deprotection. ROCK2-@4 and ROCK2-@15 plots are coloured in blue and green, respectively. Peptides belonging to regions of interest are underlined in different colours: glycine-rich loop (red), αC-helix (green), gatekeeper (yellow), hinge (purple), catalytic loop (blue), activation loop (pink) and C-tail (grey). (b) Projection of HDX data obtained for derivative 15 on the X-ray structure of ROCK2 in complex with compound **4**. When multiple $\sum\Delta$DU values were associated to one single residue, only the maximum one was considered in the analysis. While cartoons are coloured according to the $\sum\Delta$DU value, from blue (−12.8) to red (5.1), the co-crystallized ligand pose was depicted with green sticks.

(Table 4). Compound **4** showed negligible kinetic solubility, high permeability and high clearance in rat microsomes, with a half-life of 18 minutes. The significant permeability value could be rationalized by considering the lipophilicity of the compound **4** (clog *D*: 5.3). Notably, the introduction of a pendant 3-(4-methylpiperazyl)propanyl chain in **15** led to a dramatic increase in solubility, probably due to the presence

Table 3 Inhibitory activities of selected analogues in PASM cell line

Compound	pEC_{50}
4	7.9 ± 0.1
6	7.3 ± 0.4
9	6.5 ± 0.4
10	6.4 ± 0.2
14	7.5 ± 0.1
15	8.3 ± 0.1

Table 4 Preliminary ADME data for compounds **4** and **15**

	Compound **4**	Compound **15**
Kinetic solubility, μM (PBS pH: 7.2)	<5	131
Mics Cli rat Thalf (min)	18	8.4
P_{app} (AB) Caco-2 (nm s^{-1})	269 (high)	56 (moderate)

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of a weakly basic piperazine (calculated main basic pK_a: 7.6) which is partially charged at physiological pH, as well as to a reduced lipophilic character (clog *P*: 4.3), that led to a slight decrease in permeability. Although compound **15** exhibited a lower lipophilicity compared to **4** (clog *P*: 4.3 and clog *D*: 3.7), the microsomal turnover in rat slightly increased, with a half-life of 8.4 minutes.

Chemistry

Compounds of Table 1 were prepared according to Schemes 1 and 2. In the first step, a non-regioselective Schmidt reaction was performed to convert 7-bromo-3,4-dihydronaphthalen-2(1*H*)-one into the two expected azepinones: 8-bromo-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (**17**, scaffold A) and 8-bromo-1,2,4,5-tetrahydro-3*H*-benzo[*c*]azepin-3-one (**18**, scaffold B). The insertion of aryl groups onto the lactam moiety of the azepinone was carried out by means of a Pd-catalyzed *N*-arylation, reacting **17** (or **18**) with the appropriate aryl iodide **19a–d** to give intermediates **3a**, **4a**, **5a**, **14a**, and **15a**. Single or multistep conversion of intermediates **3a**, **4a**, **14a**, and **15a** to compounds **3**, **4**, **6–11**, **14–15** is reported in Scheme 2. Preparation of intermediates **3b**, **4b**, **7b**, **14b**, and **15b** was performed using a two-step/one-pot process; first the Pd-catalyzed borylation of bromo intermediates **3a**, **4a**, **14a**, and **15a**, then a Suzuki–Miyaura cross-coupling with the corresponding SEM-protected heteroaryl bromides **20a–b**.

Cleavage of the SEM group was performed under acidic conditions, followed by treatment with ammonia to completely remove any residual aldehyde adduct on the pyrrole nitrogen of the pyrrolopyridine moiety. SEM cleavage from **4b**, **7b**, **14b**, **15b** and **3b** led respectively to compounds **4**, **7**, **14**, **15** and intermediate **3c**. Selective sulfonylation of the aniline moiety of intermediate **3c** using mesyl chloride afforded compound **3**. Compounds **6**, **8**, and **9** were prepared similarly to intermediate **3b** starting from **4a** and a suitable heteroaryl halide **21**, **22a**, or **22b**.

Compound **5** (isomer of compound **4** based on scaffold B) was prepared using the same synthetic sequence as used for compound **4** but replacing intermediate **4a** with intermediate **5a**.

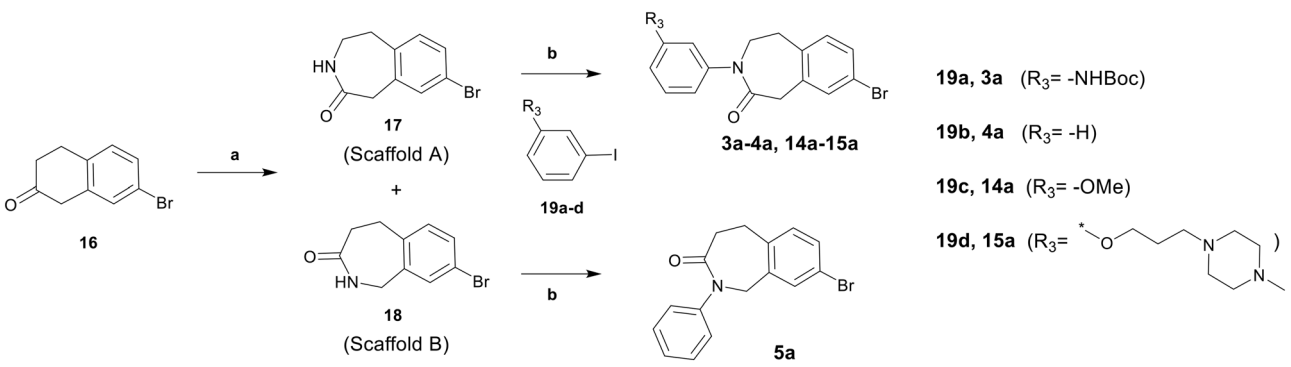
The preparation of compounds **10** and **11** was still based on Suzuki–Miyaura chemistry but revised by exploring the umpolung reactivity of intermediate **4a**. Reaction of intermediate **4a** with the corresponding boryl derivatives **23a** and **23b** led to intermediate **10a** and compound **11**, respectively. In the preparation of intermediate **10a**, THP protection of the pyrazolic NH was considered necessary to successfully perform a Suzuki–Miyaura cross-coupling. THP removal from intermediate **10a** to give compound **10** was performed by treatment with hydrochloric acid.

Finally, the syntheses of compounds **12** and **13** is depicted in Scheme 3. Introduction of aliphatic groups on the NH lactam of the azepinone moiety was approached using two different strategies. For the preparation of compound **12**, the first step was a two-step/one-pot Pd-catalysed borylation of **17** followed by Suzuki cross-coupling with heteroaryl bromide **20a** to give intermediate **12a**. The latter intermediate was easily alkylated on the lactam NH with methyl iodide in the presence of sodium hydride to give intermediate **12b**. Removal of the SEM group from intermediate **12b** afforded compound **12**.

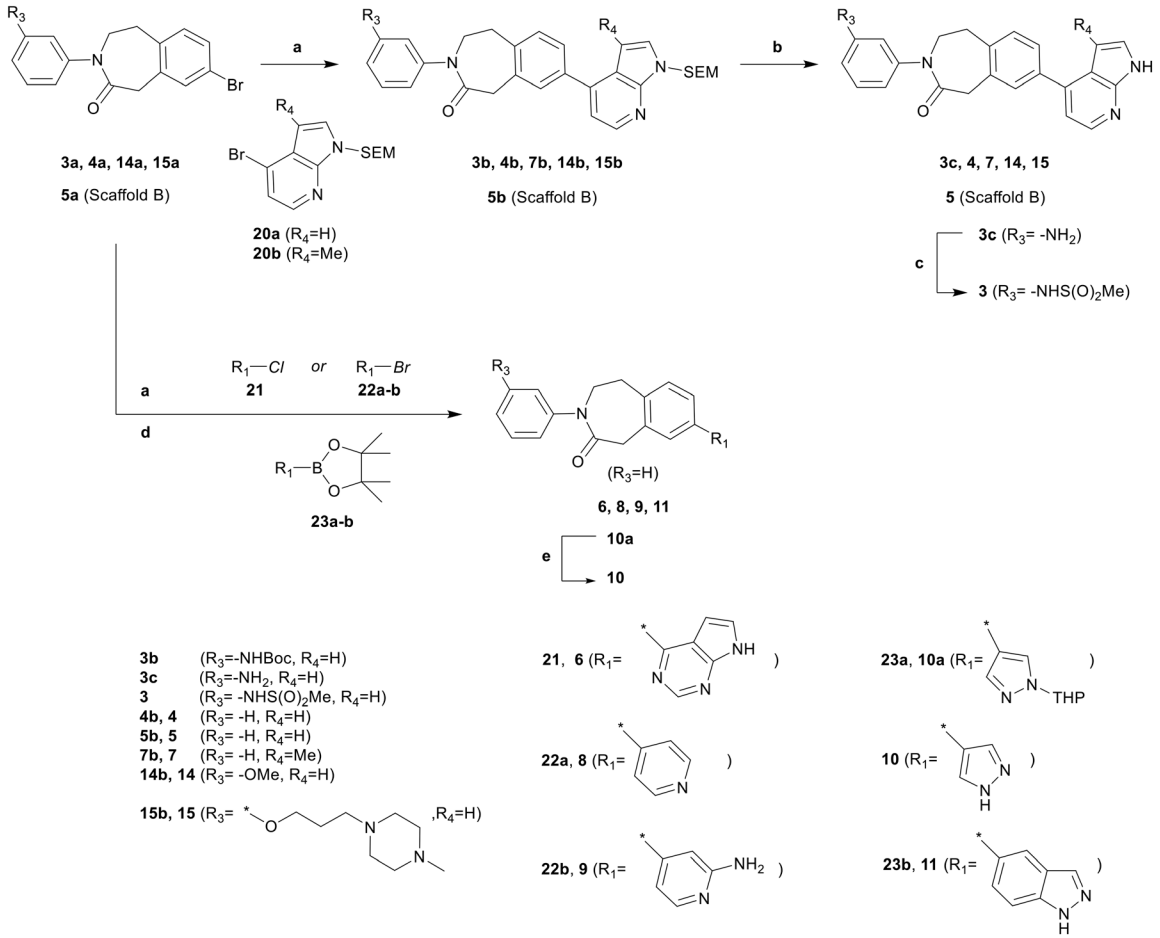
For the preparation of compound **13**, unfortunately alkylation of **12a** with 4-(2-chloroethyl)morpholine was unsuccessful and pointed out the need to tweak the synthetic sequence previously used for compound **12**. At this point, we looked at different reaction conditions to alkylate **17** with 4-(2-chloroethyl)morpholine. Despite being low yielding (yield 21%) and requiring long reaction times (two days), alkylation of **17** with 4-(2-chloroethyl)morpholine was possible in acetonitrile using a mixture of potassium and caesium carbonate with $TBAHSO_4$, allowing the preparation of intermediate **13a**. From intermediate **13a**, a two-step/one-pot Pd-catalysed borylation Suzuki–Miyaura cross-coupling with **20a** led to intermediate **13b**. By analogy with what was described previously, removal of the SEM group from **13b** led to compound **13**.

Conclusion

In summary, we have successfully exploited a combined virtual screening protocol to identify potent azaindole-based

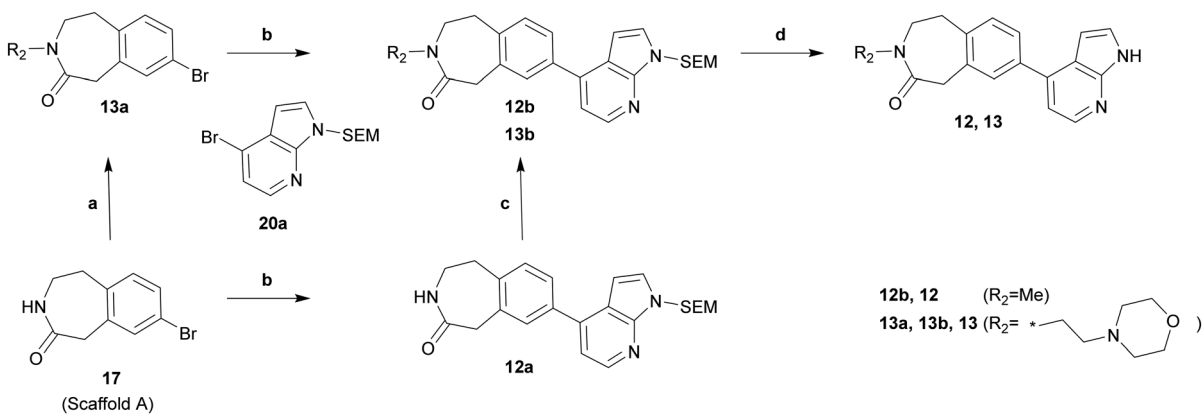


Scheme 1 Synthesis of intermediates **3a**, **4a**, **5a**, **14a** and **15a**. Conditions: (a) NaN_3, conc. H_2SO_4, toluene, 0 °C to RT, overnight; (b) Xantphos, $Pd_2(dba)_3$, Cs_2CO_3, dioxane, 100 °C, overnight.



Scheme 2 Synthesis of compounds **3**–**4**, **6**–**11** and **14**–**15**. Conditions: (a) bis(pinacolato)diboron, KOAc, Pd(dppf)Cl_2, dioxane, 105 °C then Cs_2CO_3, water and intermediates **20a** or **20b**; (b) TFA, DCM, RT then evaporation and treatment with aq. 25% NH_3, dioxane, RT, overnight; (c) mesyl chloride, TEA, DCM, 0 °C to RT; (d) Pd(dppf)Cl_2, Cs_2CO_3, water/dioxane, 105 °C, 18 h; (e) 7 M HCl in MeOH, DCM, RT, 6 h.

ROCK inhibitors bearing a novel benzoazepinone scaffold. A two-step design strategy, aimed at optimizing the hinge-binding group and the terminal aromatic moiety interacting with the glycine-rich loop, was pursued by leveraging previous SARs available for congeneric series and crystallographic information. Introduction of a flexible and positively charged side chain as in **15** represented a viable strategy to conjugate high biochemical and cellular



Scheme 3 Synthesis of compounds **12** and **13**. Conditions: (a) 4-(2-chloroethyl)morpholine, K_2CO_3, Cs_2CO_3, $TBAHSO_4$, MeCN, reflux, 2 days; (b) bis(pinacolato)diboron, KOAc, Pd(dppf)Cl_2, dioxane, 105 °C then Cs_2CO_3, water and intermediate **20a**; (c) iodomethane, NaH (60% in paraffin oil), DMF, 0 °C to RT; (d) TFA, DCM, RT then evaporation and treatment with aq. 25% NH_3, dioxane, RT, overnight.

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potencies, significant selectivity over PKA, as well as modulation of solubility and permeability. Compounds **4** and **15** showed a high metabolic turnover (in rat), suggesting that this property could be intrinsically associated with the benzoazepinone scaffold and, therefore, shared by all compounds within this series. Giving its *in vitro* metabolic profile, this series might be considered as a promising starting point for the optimization of compounds for topical/local delivery, where a high clearance by design is deemed favourable to reduce *in vivo* systemic exposure and increase the therapeutic index.

Experimental

Chemistry general procedures

All chemicals, reagents, and solvents were purchased from commercially available sources as reagent-grade products and used without further purification. Yields reported are for purified products and might not be optimized. Compounds and intermediates were analysed by ^{1}H nuclear magnetic resonance (NMR) and/or LC-MS. The reactions were monitored using analytical TLC plates (Merck, cat no. 1.05715 or equivalent) and analysed by ultraviolet light at 254 and 280 nm. Chemical reactions were purified by medium-pressure liquid chromatography. PTLC (preparative thin layer chromatography) was performed on 20 × 20 cm glass plates coated with 0.5 mm of silica gel (particle size 60 pm). After resolution, the desired band was recovered by scraping the adsorbent and elution with a strong solvent such as MeOH. NMR spectra were recorded at 300 MHz using a Bruker Fourier 300 or at 400 MHz using either a Bruker Avance 400 or a Bruker Avance III 400. Chemical shifts (δ) were reported in parts per million and consistent with chemical structure. UPLC methods are reported in ESI.†

LC-MS methods

Dionex UHPLC Ultimate 3000 with DAD detector/Thermo Scientific ISQ EC (or MSQ) mass spectrometer; column: Kinetex® 2.6 μm XB C18 (4.6 × 50 mm), 110A; flow: 1 mL min^{-1}; temperature: 25 °C; eluent A: 0.1% (v/v) HCOOH in water, eluent B: 0.1% (v/v) HCOOH in acetonitrile; (method A) gradient: 0″1 min at 95/5 A/B, 1″4.75 min from 95/5 to 20/80 A/B, 4.75″5.25 min at 20/80 A/B, 5.25″6.0 min 95/5 A/B; (method B) gradient: 0″3.35 min from 90/10 to 40/60 A/B, 3.35″3.75 min at 40/60 A/B, 3.75″3.90 min from 40/60 to 5/95 A/B, 3.90″4.75 min at 5/95 A/B, 4.75″5.0 min from 5/95 to 90/10 A/B; (method C) gradient: 0″3.35 min from 80/20 to 20/80 A/B, 3.35″3.75 min at 20/80 A/B, 3.75″3.90 min from 20/80 to 5/95 A/B, 3.90″4.75 min at 5/95 A/B, 4.75″5.00 min at 80/20 A/B. (Method D) gradient: 0″3.35 min from 90/10 to 60/40 A/B, 3.35″3.75 min at 60/40 A/B, 3.75″3.90 min from 60/40 to 5/95 A/B, 3.90″4.75 min at 5/95 A/B, 4.75″5.00 min at 90/10 A/B. UPLC + Waters DAD + Waters SQD2, single quadrupole UPLC-MS; column: 1.8 mm Acquity UPLC HSS C18 (100 × 2.1 mm); flow: 0.4 mL min^{-1}; temperature: 40 °C; eluent A: 0.1% (v/v) HCOOH in water, eluent B: 0.1% (v/v) HCOOH in acetonitrile; (method E) gradient: 0″0.4 min at 95/5 A/B, 0.4″6.0 min from 95/5 to 5/95 A/B, 6.0″6.8 min at 9/95 A/B, 6.8″7.0 min 95/5 A/B. Acquity UPLC (binary pump/PDA detector) + ZQ mass spectrometer; column: 1.7 μm Acquity UPLC BEH C18, (100 × 2.1 mm); flow: 0.4 mL min^{-1}: temperature 40 °C; eluent A: 0.1% (v/v) HCOOH in water, eluent B: 0.1% (v/v) HCOOH in acetonitrile; (method F) gradient: 0″0.4 min at 95/5 A/B, 0.4″6.0 min from 95/5 to 5/95 A/B, 6.0″6.8 min at 9/95 A/B, 6.8″7.0 min 95/5 A/B.

N-(3-(2-Oxo-8-(1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-1,2,4,5-tetrahydro-3*H*-benzo[*d*]azepin-3-yl)phenyl)methanesulfonamide (3)

To an ice cooled mixture of intermediate **3c** (100 mg, 0.27 mmol) and TEA (0.075 mL, 0.54 mmol) in DCM (1 mL), mesyl chloride (0.032 mL, 0.324 mmol) was added. The reaction mixture was allowed to reach RT and then stirred overnight. The reaction mixture was evaporated to dryness and redissolved in pyridine (1 mL) prior to the addition of a second equivalent of mesyl chloride, then further stirred at RT. The reaction mixture was evaporated to dryness and the residue partitioned between DCM and water. The organic layer was evaporated, and the residue purified by PTLC eluting with 5% v/v of MeOH in DCM to give the title compound (48 mg, yield: 40%). ^{1}H NMR (300 MHz, DMSO-d_6): δ ppm 3.01 (s, 3 H), 3.29 (br t, J = 6.0 Hz, 2 H), 4.13–4.19 (m, 2 H), 4.22 (t, J = 6.0 Hz, 2 H), 6.61 (dd, J = 3.4, 1.7 Hz, 1 H), 6.99–7.06 (m, 1 H), 7.07–7.12 (m, 1 H), 7.13–7.22 (m, 2 H), 7.29–7.41 (m, 2 H), 7.51–7.68 (m, 3 H), 8.28 (d, J = 5.0 Hz, 1 H), 9.81 (s, 1 H), 11.79 (br s, 1 H). LCMS: t_R 3.5 min (method A), purity 96.9%; MS (ESI+): 447.2 $[M + H]^+$.

3-(3-Aminophenyl)-8-(1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (3c)

Intermediate **3b** (1.18 g, 1.97 mmol) was dissolved in a mixture of DCM (11.8 mL) and TFA (5.90 mL) and stirred for 3 h at RT. The reaction mixture was evaporated to dryness, then the residue was taken into dioxane (10 mL) and treated with 25% aq NH_3 (5 mL) at RT. The reaction mixture was stirred overnight and then diluted with water (10 mL) and extracted with DCM (3 × 70 mL). The combined organic layers were dried over $MgSO_4$, filtered, and concentrated *in vacuo*. The crude material was chromatographed on silica gel by gradient elution with MeOH in DCM (0 to 5%) to give the title compound (0.70 g, yield: 96%). ^{1}H NMR (300 MHz, DMSO-d_6) δ ppm 3.20–3.30 (m, 2 H), 4.05–4.21 (m, 4 H), 5.07–5.21 (m, 2 H), 6.37–6.49 (m, 3 H), 6.61 (dd, J = 3.4, 1.7 Hz, 1 H), 7.01 (t, J = 7.98 Hz, 1 H), 7.19 (d, J = 5.0 Hz, 1 H), 7.36 (d, J = 7.89 Hz, 1 H), 7.51–7.67 (m, 3 H), 8.28 (d, J = 5.0 Hz, 1 H), 11.79 (br s, 1 H). MS (ESI+): 369.2 $[M + H]^+$.

The method used for preparation of intermediate **3c** was applied for the synthesis of compounds **4**, **5**, **7**, **12**, **13**, **14**, **15**.

3-Phenyl-8-(1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[d]azepin-2-one (4)

Prepared from **4b** (29 mg, yield: 78%). ^{1}H NMR (400 MHz, DMSO-d_6): δ ppm 3.30 (m, 2 H), 4.18 (s, 2 H), 4.22 (m, 2 H),

6.61 (d, J = 3.7 Hz, 1 H), 7.19 (d, J = 4.84 Hz, 1 H), 7.22–7.27 (m, 1 H), 7.28–7.32 (m, 2 H), 7.34–7.42 (m, 3 H), 7.54 (d, J = 3.4 Hz, 1 H), 7.60 (m, 1 H), 7.64 (dd, J = 1.7 Hz, J = 8.3 Hz, 1 H), 8.28 (d, J = 4.9 Hz, 1 H), 11.78 (s, 1 H). LCMS: t_R 3.6 min (method E), purity 99.8%; MS (ESI+): 354.2 $[M + H]^+$.

2-Phenyl-8-(1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-1,2,4,5-tetrahydro-3*H*-benzo[*c*]azepin-3-one (5)

Prepared from **5b** (13 mg, yield: 38%). ^{1}H NMR (400 MHz, DMSO-d_6): δ ppm 3.09 (m, 2 H), 3.28 (m, 2 H), 5.11, (s, 2 H), 6.59 (d, J = 4.0 Hz, 1 H), 7.18–7.23 (m, 4 H), 7.32–7.38 (m, 2 H), 7.41 (d, J = 8.0 Hz, 1 H), 7.53 (d, J = 3.9 Hz, 1H), 7.64–7.71 (m, 2H), 8.26 (d, J = 5.2 Hz, 1H), 11.78 (s, 1H). LCMS: t_R 3.3 min (method E), purity 98.2%; MS (ESI+): 354.2 $[M + H]^+$.

8-(3-Methyl-1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-3-phenyl-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (7)

Prepared from **7b** (0.8 mg, yield: 3%). ^{1}H NMR (400 MHz, DMSO-d_6): δ ppm 1.91 (s, 3 H), 4.15 (s, 2 H), 4.23 (m, 2 H), 6.88 (d, J = 4.8 Hz, 1 H), 7.22–7.32 (m, 7 H), 7.37–7.42 (m, 2 H), 8.19 (d, J = 4.8 Hz, 1 H), 11.44 (s, 1H), CH_2 under solvent. LCMS: t_R 3.7 min (method E), purity 98.1%; MS (ESI+): 368.3 $[M + H]^+$.

3-Methyl-8-(1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (12)

Prepared from **12b** (3.5 mg, yield: 22%). ^{1}H NMR (300 MHz, CD_3OD): δ ppm 3.04 (s, 3 H), 3.18–3.26 (m, 2 H), 3.78–3.95 (m, 2 H), 3.95–4.04 (m, 2 H), 6.68 (d, J = 3.5 Hz, 1 H), 7.12–7.25 (m, 1 H), 7.31 (d, J = 7.9 Hz, 1 H), 7.40–7.46 (m, 1 H), 7.48–7.54 (m, 1 H), 7.55–7.62 (m, 1 H), 7.99–8.13 (m, 1 H), 8.13–8.47 (br m, 1 H). LCMS: t_R 2.07 min (method B), purity 97.2%; MS (ESI+): 292.0 $[M + H]^+$.

3-(2-Morpholinoethyl)-8-(1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (13)

Prepared from **13b** (22 mg, yield: 39%). ^{1}H NMR (300 MHz, CD_3OD): δ ppm 2.59–2.85 (m, 5 H), 3.19–3.27 (m, 2 H), 3.60–3.65 (m, 1 H), 3.63 (s, 2 H), 3.65–3.72 (m, 5 H), 3.82–3.94 (m, 2 H), 4.00 (s, 2 H), 6.57–6.87 (m, 1 H), 7.03–7.25 (m, 1 H), 7.28 (d, J = 7.9 Hz, 1 H), 7.41 (d, J = 3.5 Hz, 1 H), 7.50 (s, 1 H), 7.53–7.66 (m, 1 H), 7.98–8.97 (m, 1 H). LCMS: t_R 2.56 min (method A), purity 99.8%; MS (ESI+): 391.0 $[M + H]^+$.

3-(3-Methoxyphenyl)-8-(1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (14)

Prepared from **14b** (34 mg, yield: 75%). ^{1}H NMR (300 MHz, DMSO-d_6): δ ppm 3.30 (br s, 2 H), 3.75 (s, 3 H), 4.10–4.28 (m, 5 H), 6.55–6.66 (m, 1 H), 6.79–6.91 (m, 3 H), 7.19 (d, J = 5.0 Hz, 1 H), 7.25–7.33 (m, 1 H), 7.33–7.42 (m, 1 H), 7.51–7.57 (m, 1 H), 7.58–7.68 (m, 2 H), 8.28 (d, J = 6.0 Hz, 1 H), 11.70–11.87 (m, 1 H). LCMS: t_R 2.19 min (method C), purity 98.8%; MS (ESI+): 384.0 $[M + H]^+$.

3-(3-(3-(4-Methylpiperazin-1-yl)propoxy)phenyl)-8-(1*H*-pyrrolo[2,3-b]pyridin-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (15)

Prepared from **15b** (33 mg, yield: 69%). ^{1}H NMR (300 MHz, DMSO-d_6): δ ppm 1.85 (quin, J = 6.8 Hz, 2 H), 2.26 (s, 3 H), 2.37–2.47 (m, 5 H), 3.22–3.42 (m, 6 H), 3.98 (t, J = 6.2 Hz, 2 H), 4.12–4.29 (m, 4 H), 6.57–6.64 (m, 1 H), 6.77–6.91 (m, 3 H), 7.19 (d, J = 5.2 Hz, 1 H), 7.24–7.32 (m, 1 H), 7.36 (d, J = 8.1 Hz, 1 H), 7.52–7.57 (m, 1 H), 7.58–7.61 (m, 1 H), 7.61–7.68 (m, 1 H), 8.28 (d, J = 5.1 Hz, 1 H), 11.8 (s, 1 H). LCMS: t_R 2.25 min (method D), purity 97.6%; MS (ESI+): 510.1 $[M + H]^+$.

***tert*-Butyl (3-(2-oxo-8-(1-((2-(trimethylsilyl)ethoxy)methyl)-1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-1,2,4,5-tetrahydro-3*H*-benzo[*d*]azepin-3-yl)phenyl)carbamate (3b)**

To a degassed mixture of intermediate **3a** (1.0 g, 2.3 mmol), bis(pinacolato)diboron (0.59 g, 2.3 mmol) and KOAc (0.59 g, 8.0 mmol) in dioxane (50 mL), Pd(dppf)Cl_2 (170 mg, 0.23 mmol) was added, and the reaction mixture heated at 105 °C for 4.5 h. After cooling to RT, water (5 mL), Cs_2CO_3 (1.5 g, 4.6 mmol) and intermediate **20a** (2.3 mmol, 750 mg) were added, and the reaction mixture was stirred at 105 °C for 5.5 h. After cooling to RT, the reaction mixture was filtered through a pad of diatomaceous earth and concentrated *in vacuo*. The crude material was chromatographed on silica gel by gradient elution with EtOAc in hexane (5% to 40%) to give the title product (1.18 g, yield: 84%). ^{1}H NMR (300 MHz, DMSO-d_6): δ ppm −0.11–−0.07 (m, 9 H), 0.80–0.89 (m, 2 H), 1.46 (s, 9 H), 3.23–3.30 (m, 2 H), 3.49–3.60 (m, 2 H), 4.13–4.24 (m, 4 H), 5.67 (s, 2 H), 6.71 (d, J = 3.7 Hz, 1 H), 6.88 (dt, J = 7.2, 1.8 Hz, 1 H), 7.21–7.32 (m, 3 H), 7.38 (d, J = 7.5 Hz, 1 H), 7.44–7.50 (m, 1 H), 7.59–7.67 (m, 2 H), 7.73 (d, J = 3.6 Hz, 1 H), 8.35 (d, J = 5.2 Hz, 1 H), 9.41 (br s, 1 H). MS (ESI$^+$): 599.8 $[M + H]^+$.

The method used for preparation of intermediate **3b** was applied for the synthesis of intermediates **4b**, **5b**, **7b**, **12a**, **13b**, **14b**, **15b** and compounds **6**, **8** and **9**.

3-Phenyl-8-(1-((2-(trimethylsilyl)ethoxy)methyl)-1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (4b)

Prepared from **4a** and **20a** (50 mg, yield: 82%). MS (ESI+): 484.2 $[M + H]^+$.

2-Phenyl-8-(1-((2-(trimethylsilyl)ethoxy)methyl)-1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-1,2,4,5-tetrahydro-3*H*-benzo[*c*]azepin-3-one (5b)

Prepared from **5a** and **20a** (47 mg, yield: 62%). MS (ESI+): 484.2 $[M + H]^+$.

8-(3-Methyl-1-((2-(trimethylsilyl)ethoxy)methyl)-1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-3-phenyl-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (7b)

Prepared from **4a** and **20b** (23 mg, yield: 37%). (ESI+): 497.7 $[M + H]^+$.

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8-(1-((2-(Trimethylsilyl)ethoxy)methyl)-1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (12a)

Prepared from (scaffold A) and **20a**. ^{1}H NMR (300 MHz, DMSO-d_6) δ ppm −0.12–−0.05 (m, 9 H), 0.78–0.88 (m, 2 H), 1.15 (s, 2 H), 3.09 (br t, J = 6.0 Hz, 2 H), 3.45–3.61 (m, 4 H), 3.87 (s, 3 H), 5.63–5.71 (m, 2 H), 6.67 (d, J = 3.7 Hz, 1 H), 7.25 (d, J = 5.0 Hz, 1 H), 7.32 (d, J = 7.9 Hz, 1 H), 7.53 (d, J = 1.5 Hz, 1 H), 7.59 (dd, J = 7.9, 1.8 Hz, 1 H), 7.71 (d, J = 3.7 Hz, 1 H), 8.33 (d, J = 5.0 Hz, 1 H). MS (ESI+): 408.8 $[M + H]^+$.

3-(2-Morpholinoethyl)-8-(1-((2-(trimethylsilyl)ethoxy)methyl)-1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (13b)

Prepared from **13a** and **20a** (72 mg, yield: 64%). ^{1}H NMR (300 MHz, DMSO-d_6): δ ppm −0.09 (s, 9 H), 0.79–0.88 (m, 2 H), 2.34–2.45 (m, 6 H), 3.42–3.58 (m, 9 H), 3.80–3.90 (m, 2 H), 3.95 (d, J = 9.5 Hz, 2 H), 4.10 (q, J = 5.3 Hz, 1 H), 5.66 (s, 2 H), 6.66 (d, J = 3.7 Hz, 1 H), 7.25 (d, J = 5.0 Hz, 1 H), 7.31 (d, J = 7.9 Hz, 1 H), 7.52 (d, J = 1.7 Hz, 1 H), 7.58 (dd, J = 7.9, 1.8 Hz, 1 H), 7.71 (d, J = 3.5 Hz, 1 H), 8.33 (d, J = 5.0 Hz, 1 H). MS (ESI+): 521.8 $[M + H]^+$.

3-(3-Methoxyphenyl)-8-(1-((2-(trimethylsilyl)ethoxy)methyl)-1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (14b)

Prepared from **14a** and **20a** (64 mg, yield: 80%). ^{1}H NMR (300 MHz, DMSO-d_6): δ ppm −0.11–−0.06 (m, 9 H), 0.80–0.90 (m, 2 H), 3.26–3.31 (m, 2 H), 3.45–3.63 (m, 2 H), 3.75 (s, 3 H), 4.10–4.29 (m, 4 H), 5.67 (s, 2 H), 6.70 (d, J = 3.7 Hz, 1 H), 6.81–6.89 (m, 3 H), 7.21–7.34 (m, 2 H), 7.38 (d, J = 8.0 Hz, 1 H), 7.56–7.69 (m, 2 H), 7.73 (d, J = 3.7 Hz, 1 H), 8.34 (d, J = 5.0 Hz, 1 H). MS (ESI+): 514.3 $[M + H]^+$.

3-(3-(3-(4-Methylpiperazin-1-yl)propoxy)phenyl)-8-(1-((2-(trimethylsilyl)ethoxy)methyl)-1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (15b)

Prepared from **15a** and **20a** (64 mg, yield: 58%). ^{1}H NMR (300 MHz, DMSO-d_6): δ ppm −0.13–−0.04 (m, 9 H), 0.79–0.89 (m, 2 H), 1.84 (quin, J = 6.7 Hz, 2 H), 2.14–2.20 (m, 3 H), 2.29–2.46 (m, 8 H), 3.23–3.39 (m, 4 H), 3.50–3.59 (m, 2 H), 3.98 (t, J = 6.3 Hz, 2 H), 4.09–4.31 (m, 4 H), 5.67 (s, 2 H), 6.70 (d, J = 3.7 Hz, 1 H), 6.78–6.87 (m, 3 H), 7.21–7.33 (m, 2 H), 7.37 (d, J = 7.9 Hz, 1 H), 7.56–7.68 (m, 2 H), 7.73 (d, J = 3.5 Hz, 1 H), 8.35 (d, J = 5.0 Hz, 1 H). MS (ESI+): 640.3 $[M + H]^+$.

3-Phenyl-8-(7*H*-pyrrolo[2,3-*d*]pyrimidin-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (6)

Prepared from **4a** and 4-chloro-7*H*-pyrrolo[2,3-*d*]pyrimidine (16 mg, yield: 31%). ^{1}H NMR (300 MHz, CD_3OD): δ ppm 3.35 (t, J = 6.1 Hz, 2 H), 4.16–4.23 (m, 4 H), 6.82 (d, J = 3.7 Hz, 1 H), 7.20–7.29 (m, 3 H), 7.32–7.41 (m, 5 H), 7.84–7.94 (m, 2 H), 8.79 (s, 1 H). LCMS: t_R 1.83 min (method C), purity 96.8%; MS (ESI+): 355.0 $[M + H]^+$.

3-Phenyl-8-(pyridin-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (8)

Prepared from **4a** and 4-bromopyridine (7 mg, yield: 13%). ^{1}H NMR (400 MHz, $CDCl_3$): δ 3.33 (m, 2 H), 4.16–4.22 (m, 4 H), 7.24–7.30 (m, 4 H), 7.37–7.43 (m, 2 H), 7.47–7.53 (m, 4 H), 8.67 (d, J = 5.7 Hz, 2 H). LCMS: t_R 2.6 min (method F), purity 99.4%; MS (ESI): 315.2 $[M + H]^+$.

8-(2-Aminopyridin-4-yl)-3-phenyl-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (9)

Prepared from **4a** and 4-bromopyridin-2-amine (27 mg, yield: 52%). ^{1}H NMR (300 MHz, DMSO-d_6): δ ppm 3.26 (br t, J = 5.8 Hz, 2 H), 4.14 (s, 2 H), 4.16–4.26 (m, 2 H), 5.95 (s, 2 H), 6.68–6.75 (m, 1 H), 6.80 (dd, J = 5.4, 1.6 Hz, 1 H), 7.21–7.32 (m, 4 H), 7.32–7.41 (m, 2 H), 7.46–7.53 (m, 2 H), 7.96 (d, J = 5.3 Hz, 1 H). LCMS: t_R 1.49 min (method C), purity 99.3%; MS (ESI+): 330.0 $[M + H]^+$.

3-Phenyl-8-(1*H*-pyrazol-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (10)

To a solution of intermediate **10a** (29.0 mg, 0.07 mmol) in DCM (0.5 mL), 7 M HCl in MeOH (0.7 mL) was added, and the mixture was stirred at RT for 6 h. The reaction mixture was evaporated, and the crude material purified by PTLC by elution with 10% MeOH in DCM to give the title product (16 mg, yield: 76%). ^{1}H NMR (300 MHz, CD_3OD): δ ppm 3.26 (t, J = 6.0 Hz, 2 H), 4.04–4.12 (m, 2 H), 4.17 (t, J = 6.0 Hz, 2 H), 7.12–7.31 (m, 4 H), 7.31–7.44 (m, 4 H), 7.85 (s, 2 H). LCMS: t_R 2.40 min (method C), purity 99.6%; MS (ESI+): 304.0 $[M + H]^+$.

3-Phenyl-8-(1-(tetrahydro-2*H*-pyran-2-yl)-1*H*-pyrazol-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (10a)

To a degassed mixture of **4a** (50.0 mg, 0.15 mmol), 1-(tetrahydro-2*H*-pyran-2-ylo)-1*H*-pyrazole-4-boronic acid pinacol ester (22.0 mg, 0.08 mmol), and Cs_2CO_3 (51.5 mg, 0.16 mmol) in dioxane (1.25 mL) and water (0.13 mL), Pd(dppf)Cl_2 (5.8 mg, 0.008 mmol) was added, and reaction mixture heated to 105 °C for 18 h. After evaporation, the crude material was purified by PTLC eluting with 50% EtOAc in hexane to give the title product (29 mg, yield: 94%). MS (ESI): 388.7 $[M + H]^+$.

8-(1*H*-Indazol-5-yl)-3-phenyl-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (11)

This compound was prepared similarly to intermediate **10a** starting from **4a** and 5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1*H*-indazole (15 mg, yield: 16%). ^{1}H NMR (300 MHz, CD_3OD): δ ppm 3.32–3.37 (m, 2 H), 4.17 (s, 2 H), 4.24 (t, J = 5.9 Hz, 2 H), 7.19–7.33 (m, 4 H), 7.36–7.48 (m, 3 H), 7.49–7.74 (m, 4 H), 7.96 (s, 1 H), 8.02–8.11 (m, 1 H). LCMS: t_R 2.93 min (method C), purity 90.1%; MS (ESI+): 353.8 $[M + H]^+$.

***tert*-Butyl (3-(8-bromo-2-oxo-1,2,4,5-tetrahydro-3*H*-benzo[*d*]azepin-3-yl)phenyl)carbamate (3a)**

To a degassed mixture of scaffold A (3.0 g, 12.5 mmol), *N*-(3-iodophenyl)carbamate (4.39 g, 18.8 mmol) and Cs_2CO_3 (5.69 g, 17.5 mmol) in dioxane (120 mL), Xantphos (540 mg, 0.099 mmol) and $Pd_2(dba)_3$ (0.57 g, 0.62 mmol) were added and the reaction mixture heated at 100 °C overnight. After cooling to RT, the reaction mixture was filtered through a pad of diatomaceous earth and concentrated *in vacuo*. The resulting crude product was chromatographed on silica gel by gradient elution with EtOAc in hexane (3% to 40%). The resulting solid was triturated in Et_2O to give the title compound (1.0 g, yield: 19%). ^{1}H NMR (300 MHz, DMSO-d_6): δ ppm 1.45 (s, 9 H), 3.08–3.21 (m, 2 H), 4.02–4.18 (m, 4 H), 6.83 (br d, J = 7.2 Hz, 1 H), 7.15 (d, J = 8.3 Hz, 1 H), 7.18–7.31 (m, 2 H), 7.33–7.45 (m, 2 H), 7.47 (s, 1 H), 9.40 (s, 1 H). MS (ESI+): 430.9/432.8 $[M + H]^+$.

The method used for preparation of intermediate **3a** was applied for the synthesis of intermediates **4a**, **5a**, **14a**, **15a**.

8-Bromo-3-phenyl-1,3,4,5-tetrahydro-2*H*-benzo[d]azepin-2-one (4a)

Prepared from **17** and iodobenzene (557 mg, yield: 42%). ^{1}H NMR (300 MHz, DMSO-d_6): δ ppm 3.17 (t, J = 5.9 Hz, 2 H), 4.07 (s, 2 H), 4.10–4.19 (m, 2 H), 7.14 (d, J = 8.3 Hz, 1 H), 7.19–7.29 (m, 3 H), 7.30–7.45 (m, 3 H), 7.48 (d, J = 2.02 Hz, 1 H). MS (ESI): 315.9/317.9 $[M + H]^+$.

8-Bromo-2-phenyl-1,2,4,5-tetrahydro-3*H*-benzo[*c*]azepin-3-one (5a)

Prepared from **18** and iodobenzene (115 mg, yield: 77%). MS (ESI): 315.9/317.9 $[M + H]^+$.

8-Bromo-3-(3-methoxyphenyl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (14a)

Prepared from **17** and 1-iodo-3-methoxybenzene (57 mg, yield: 40%). ^{1}H NMR (300 MHz, DMSO-d_6): δ ppm 3.16 (br t, J = 5.9 Hz, 2 H), 3.74 (s, 3 H), 4.06 (s, 2 H), 4.13 (br t, J = 5.9 Hz, 2 H), 6.77–6.85 (m, 3 H), 7.14 (d, J = 8.3 Hz, 1 H), 7.21–7.33 (m, 1 H), 7.40 (dd, J = 8.3, 2.2 Hz, 1 H), 7.47 (d, J = 2.2 Hz, 1 H). MS (ESI): 346/348.1 $[M + H]^+$.

8-Bromo-3-(3-(3-(4-methylpiperazin-1-yl)propoxy)phenyl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (15a)

Prepared from **17** and **19d** (73 mg, yield: 37%). ^{1}H NMR (300 MHz, DMSO-d_6): δ ppm 1.83 (br t, J = 6.9 Hz, 2 H), 2.10–2.19 (m, 3 H), 2.26–2.43 (m, 8 H), 3.05–3.26 (m, 4 H), 3.96 (t, J = 6.3 Hz, 2 H), 4.05 (s, 2 H), 4.12 (t, J = 6.3 Hz, 2 H), 6.76–6.86 (m, 3 H), 7.08–7.19 (m, 1 H), 7.26 (t, J = 8.3 Hz, 1 H), 7.34–7.53 (m, 2 H). MS (ESI): 472/474.1 $[M + H]^+$.

3-Methyl-8-(1-((2-(trimethylsilyl)ethoxy)methyl)-1*H*-pyrrolo[2,3-*b*]pyridin-4-yl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (12b)

To a solution of intermediate **12a** (199 mg, 0.25 mmol) in DMF (3 mL), NaH 60% in paraffin oil (10.8 mg, 0.27 mmol) was added in one portion and the resulting mixture stirred at RT for 20 min. The reaction mixture was cooled to 0 °C before the addition of iodomethane (0.023 mL, 0.37 mmol), then warmed to RT and stirred overnight. After evaporation *in vacuo*, the resulting mixture was concentrated and the crude was purified by PTLC eluting with 30–40% EtOAc in DCM to give the title product (20 mg, yield: 20%). MS (ESI+): 422.5 $[M + H]^+$.

8-Bromo-3-(2-morpholinoethyl)-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (13a)

To a mixture of **17** (100 mg, 0.42 mmol), 4-(2-chloroethyl) morpholine hydrochloride (78 mg, 0.42 mmol), K_2CO_3 (173 mg, 1.26 mmol), Cs_2CO_3 (271 mg, 0.84 mmol) in MeCN (2 mL), a catalytic amount of $TBAHSO_4$ was added and reaction mixture was refluxed for 2 days. After evaporation *in vacuo*, the residue was treated with 20% MeOH in DCM and filtered through a pad of diatomaceous earth. The filtrate was concentrated, and the resulting crude chromatographed on silica gel by gradient elution with MeOH in DCM (0 to 6%). Fractions containing the product were evaporated and purified again by PTLC with 5% MeOH in DCM to give the desired product (32 mg, yield: 22%). ^{1}H NMR (300 MHz, $CDCl_3$): δ ppm 2.42–2.49 (m, 4 H), 2.49–2.53 (m, 2 H), 3.08–3.15 (m, 2 H), 3.55 (t, J = 6.7 Hz, 2 H), 3.59–3.68 (m, 4 H), 3.70–3.77 (m, 2 H), 3.84 (s, 2 H), 6.95 (d, J = 8.3 Hz, 1 H), 7.25–7.31 (m, 2 H).

8-Bromo-1,3,4,5-tetrahydro-2*H*-benzo[*d*]azepin-2-one (17)

To a stirred solution of 7-bromo-3,4-dihydro-1*H*-naphthalen-2-one (24 g, 0.11 mol) in toluene (480 mL), sodium azide (7.6 g, 0.12 mol) was added, and the resulting mixture cooled to 0 °C prior to the dropwise addition of concentrated sulfuric acid (97.3 mL, 1.8 mol). The reaction mixture was allowed to warm to RT and stirred overnight. The reaction mixture was diluted with EtOAc and washed with water and saturated aqueous NaCl. The aqueous phase was back-extracted with EtOAc, and the combined organic layers dried over Na_2SO_4 and solvents evaporated *in vacuo*. The resulting crude was chromatographed on silica gel by eluting with MeCN (2% EtOH) in DCM to give the title product (10.4 g, yield: 41%). ^{1}H NMR (300 MHz, DMSO-d_6): δ ppm 2.91–3.00 (m, 2 H), 3.39–3.46 (m, 2 H), 3.75 (s, 2 H), 7.09 (d, J = 8.1 Hz, 1 H), 7.33–7.42 (m, 2 H), 7.63 (br s, 1 H). MS (ESI): 240.0/242.0 $[M + H]^+$.

8-Bromo-1,2,4,5-tetrahydro-3*H*-benzo[*c*]azepin-3-one (18)

To a solution of 7-bromo-3,4-dihydronaphthalen-2(1*H*)-one (1.0 g, 0.0044 mol) in toluene (20 mL) was added sodium azide (0.58 g, 0.0089 mol) and the mixture was cooled to 0 °C.

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Sulfuric acid (4.0 mL, 0.075 mol) was added slowly and the reaction stirred at RT overnight. The mixture was partitioned between EtOAc and water. The organic was separated, and the aqueous layer was twice further extracted with EtOAc. The combined organics were washed with water and then brine, dried over Na_2SO_4 and evaporated. The regioisomeric mixture was separated by chromatography on a 40 g Si cartridge eluting with MeCN (1% EtOH)/DCM (5–70% gradient). Pure scaffold A was obtained as a pale brown solid (257 mg, yield: 24%) and scaffold B as a grey/brown solid (253 mg, yield: 24%), with mixed fractions totalling 333 mg also being recovered. For scaffold B: ^{1}H NMR (400 MHz, DMSO-d_6): δ ppm 2.58–2.65 (m, 2 H), 2.93–3.00 (m, 2 H), 4.25 (d, J = 7.3 Hz, 2 H), 7.15 (d, J = 10.6 Hz, 1 H), 7.38–7.45 (m, 2 H), 7.93 (t, J = 7.3 Hz, 1 H). MS (ESI): 240.1/242.1 $[M + H]^+$.

1-(3-(3-Iodophenoxy)propyl)-4-methylpiperazine (19d)

Step 1 (**19d-a**). A mixture of 3-iodophenol (1.0 g, 0.0045 mol), 1,3-dibromopropane (0.92 mL, 0.0091 mol) and K_2CO_3 (1.26 g, 0.0091 mmol) in DMF (6 mL) was stirred at 60 °C for 2 h. The resulting mixture was cooled and quenched into cold water (50 mL) and extracted with EtOAc (2 × 30 mL). The combined organic layers were washed with aq. sat. NaCl, dried over $MgSO_4$ and concentrated *in vacuo*. The resulting crude was chromatographed on silica gel eluting with hexanes to give the title intermediate (0.89 g, yield: 57%). ^{1}H NMR (300 MHz, $CDCl_3$): δ ppm 2.32 (quin, J = 6.1 Hz, 2 H), 3.60 (t, J = 6.3 Hz, 2 H), 4.09 (t, J = 5.8 Hz, 2 H), 6.84–6.94 (m, 1 H), 6.94–7.05 (m, 1 H), 7.25–7.33 (m, 2 H). Step 2. To a mixture of **19d-a** (0.89 g, 0.026 mmol), *N*-methylpiperazine (0.29 mL, 0.0026 mol) in MeCN (8.9 mL), K_2CO_3 (0.72 g, 0.0052 mol) was added followed by a catalytical amount of NaI. The resulting mixture was heated at 60 °C for 3 h. The solvent was evaporated, and the residue partitioned between water (30 mL) and DCM (25 mL). The aqueous layer was further extracted with DCM (25 mL), and the combined organics were dried over $MgSO_4$ and concentrated *in vacuo*. The crude product was chromatographed on silica gel by gradient elution with 0–10% MeOH in DCM to give the title product (0.65 g, yield: 69%). ^{1}H NMR (300 MHz, $CDCl_3$): δ ppm 1.91–2.03 (m, 2 H), 2.35 (s, 3 H), 2.46–2.65 (m, 10 H), 3.99 (t, J = 6.3 Hz, 2 H), 6.82–6.91 (m, 1 H), 6.99 (t, J = 8.0 Hz, 1 H), 7.23–7.31 (m, 2 H). MS (ESI+): 361.1 $[M + H]^+$.

4-Bromo-1-((2-(trimethylsilyl)ethoxy)methyl)-1*H*-pyrrolo[2,3-*b*]pyridine (20a)

To a 0 °C cooled solution of 4-bromo-1*H*-pyrrolo[2,3-*b*]pyridine (5.0 g, 0.025 mol) and diisopropylethylamine (2.32 mL, 0.033 mol) in DCM (100 mL), 2-(trimethylsilyl)ethoxymethyl chloride (5.85 mL, 0.033 mol) was added and reaction mixture stirred at RT overnight. The reaction mixture was quenched by the addition of water and the aqueous layer was extracted twice with DCM. The combined organic layers were dried over Na_2SO_4 and concentrated *in vacuo*. The resulting crude was chromatographed on silica gel by eluting with MeOH in DCM (2–4%) to give the title product (7.74 g, yield: 93%). ^{1}H NMR (300 MHz, DMSO-d_6): δ ppm −0.18–−0.07 (m, 9 H), 0.76–0.86 (m, 2 H), 3.44–3.56 (m, 2 H), 5.63 (s, 2 H), 6.52 (d, J = 3.7 Hz, 1 H), 7.42 (d, J = 5.1 Hz, 1 H), 7.78 (d, J = 3.5 Hz, 1 H), 8.15 (d, J = 5.1 Hz, 1 H).

4-Bromo-3-methyl-1-((2-(trimethylsilyl)ethoxy)methyl)-1*H*-pyrrolo[2,3-*b*]pyridine (20b)

Prepared similarly to intermediate **20a** starting from 4-bromo-3-methyl-1*H*-pyrrolo[2,3-*b*]pyridine (8.2 g, yield: 82%). ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.08 (d, 1H, J = 5.6 Hz), 7.35 (d, 1H, J = 5.6 Hz), 7.21 (d, 1H, J = 1.0 Hz), 5.76 (s, 2H), 3.58 (t, 2H, J = 8.0 Hz), 2.53 (d, 3H, J = 1.0 Hz), 0.92 (t, 2H, J = 8.0 Hz), −0.05 (s, 9H) MS (ESI): 341.1/343.1 $[M + H]^+$.

Molecular modelling

Molecular modeling calculations were performed with the Schrödinger software suite 2020.2. ROCK1 X-ray structure was prepared with the Protein Preparation Wizard tool.[42] Ligand molecules were prepared using LigPrep 5.4 (ref. 43) whereas docking studies were carried out with Glide 8.7 (ref. 44 and 45) using the SP scoring function. Default settings were used, unless stated otherwise.

Protein preparation and docking grids

The X-ray structure of human ROCK1 co-crystallized with an azaindolyl-thiazole inhibitor was retrieved from the Protein Data Bank (PDB: 5KKS) and subsequently processed using the Protein Preparation Wizard tool. Briefly, missing hydrogen atoms were added to the protein–ligand complex and protein C- and N-termini were capped with acetyl and methylamino groups, respectively. Missing side chains and loops were reconstructed with Prime 6.0,[46,47] while the hydrogen bond (H-bond) network was optimized by sampling the orientation of polar side chains as well as the tautomerization states of histidine residues. All hydrogen atoms were then energy minimized to further refine the optimized H-bond network. A final energy minimization was conducted to relieve major steric clashes, retaining all hydrogens free to move while restraining all heavy atoms with a harmonic potential of 25 kcal mol^{-1} Å^{-2}. All water molecules were removed from the final energy-minimized structure. Chain A was then extracted from the dimeric assembly and selected as the reference unit for the subsequent virtual screening campaign. Docking grids were built with Glide and centered on the co-crystallized ligand molecule, retaining default values for enclosing and bounding box dimensions, as well as for the van der Waals scaling factor.

Dataset preparation

The core-hopping applications of Schrodinger rely on different SQLite databases of linkers. In these databases,

each linker carries information about the original molecule from which it was extracted, including its SMILES string. Accordingly, SMILES codes of all original molecules included in the largest SQLite database of linkers embedded in the Schrodinger Suite (core_library_2014.1-86640.sqlite) were retrieved and prepared with the LigPrep utility, retaining the original ionization state and preserving the chirality of defined stereocenters. Each resulting molecule was then fragmented with the corefinder tool of Schrodinger, which uses pre-defined disconnection rules to identify suitable cores to be used in core-hopping calculations. Each core was then incorporated into the original co-crystallized ROCK inhibitor devoid of its central moiety. Specifically, the 2-acetylamino-thiazole portion of the co-crystallized compound was deleted and the new exit vectors for linker placement were defined on the two disconnected branches, *i.e.* position 4 on the azaindole hinge binder and the original attachment point in *meta* position on the *N*-phenylmethanesulfonamide. To systematically explore the chemical space of the linkers, all identified cores having two or more disconnections were attached to the new exit vectors defined on the terminal branches. During this phase, all possible linking combinations, generated by joining each disconnection on the core scaffold to each branch, were included in the final compound list (Fig. S1†). This procedure yielded 131 266 molecules.

Virtual screening campaign

To evaluate the ability of each newly-generated compound to retain a binding-competent geometry, a shape screening was conducted with Phase 4.9 (ref. 48) on the entire dataset, using the co-crystallized ligand structure as template. Specifically, each ligand was treated as a set of pharmacophoric sites, identified according to the Phase feature definitions, and represented as hard spheres. Once features were created, each molecule of the database was flexibly aligned to the reference molecule using the pharmacophore-derived spheres as aligning points. Volume overlap scores are then computed between sites of the same type. A single shape similarity value, proportional to the total volume overlap, is finally calculated for each ligand molecule. To better evaluate the quality of the shape-based alignment, the root mean square deviation (RMSD) value of the atoms composing the branches in each aligned compound was calculated with respect to the reference co-crystallized ligand structure. Those compounds showing a RMSD value equal or lower than 2 Å were retained and passed to the subsequent stage.

The selected set of aligned compounds was then docked in the ROCK1 active site using a customized protocol. Specifically, the pre-placed ligand poses were only energy-minimized in the field of the protein, without imposing any conformational search during the docking simulation. The "Standard Precision" mode of Glide and the OPLS3e force field were used to conduct all docking calculations. The Docking Score of Glide was then used to rank the compounds.

The 20 best-scored docking solutions were subsequently re-scored applying the MM-GBSA approach implemented in Prime 6.0. Briefly, the selected poses were energy-minimized using the local optimization feature in Prime, whereas all protein atoms were retained frozen. Energies were calculated using the OPLS3e force field and the VSGB2.1 continuum solvent model.[49] The free energy of binding (ΔG_{Bind}) was then estimated according to the following equation:

$$\Delta G = E_{complex} - E_{ligand} - E_{protein}$$

By default, corrections terms for ligand strain energy were included in the calculation of binding free-energy.

Protein expression and purification

Catalytic ROCK2 domain was expressed as fusion protein N-His_6_ZB_TEV_ROCK2(19-417) in High5 insect cells. Purification was performed by affinity on Ni-NTA agarose resin (QIAGEN), followed by TEV cleavage and further purification by anion exchange chromatography (HiTrap Q HP 5 mL, GE Healthcare) and size exclusion chromatography (HiLoad 16/600 Superdex 200, GE Healthcare), leading to a highly pure protein (>95%) stored at −80 °C in 50 mM Hepes pH 7.5, 150 mM NaCl, 2 mM DTT.

HDX-MS analysis

Prior to HDX, ROCK2 was diluted at a concentration of 10 μM in buffer E containing 10 mM phosphate at pH 7.40, either in the absence or in the presence of 30 μM inhibitor and stored refrigerated at 2 °C in the LEAP PAL automation (Trajan). The exchange reaction was initiated by diluting the samples 15.8-fold with buffer L containing 10 mM phosphate in D_2O, pD 7.40. Labeling reaction was carried out at 25 °C and then quenched at various time points (10 s, 30 s, 5 min, 10 min, 30 min, 60 min) by mixing sample 1 : 1 with ice-cold buffer Q containing 100 mM phosphate, pH 2.66. Quenched samples were kept at 2 °C for 3 minutes prior to injection for online digestion and LC-MS analysis. The HDX-MS data presented in this paper is the average of at least three technical replicate measurements for each time point.

Liquid chromatography and mass spectrometry

Quenched samples were loaded onto a cooled (0 °C) nanoACQUITY UPLC system (Waters) with HDX technology for digestion and LC separation. Samples were first subjected to online proteolysis at 20 °C through enzymate pepsin column, 300 Å, 5 μm, 2.1 × 30 mm (Waters), followed by desalting on a C18 trap column (Acquity UPLC BEH C18 VanGuard Pre-column, 130 Å, 1.7 μm, 2.1 × 5 mm, Waters) for 2.5 min at a flow rate of 200 μL min^{-1} of solvent A (0.1% formic acid in water). The peptides were then separated by reversed-phase chromatography over a C18 analytical column (Acquity UPLC BEH C18 column, 130 Å, 1.7 μm, 1 × 100 mm,

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Waters) using a 8–40% linear gradient of solvent B (0.1% formic acid in acetonitrile) over 8 min at a flow rate of 40 μL min^{-1}. Subsequently, peptides were analyzed on a hybrid Q-TOF SYNAPT G2-S mass spectrometer (Waters) with electrospray ionization in high resolution positive ion mode and lock-mass correction using Leu-Enk.

HDX-MS data evaluation and statistical analysis

Peptides were identified from non-deuterated samples using ProteinLynx Global Server 3.0 (PLGS) (Waters). Identified peptides were further filtered using DynamX 3.0 (Waters) so that only peptides with mass error for the precursor ion <5 ppm and at least three fragmentation products were accepted. Furthermore, peptides had to be identified in 75% of the non-deuterated runs. The deuterium uptake was determined for all identified peptides using DynamX 3.0 (Waters). All peptide assignments were manually verified and noisy and overlapping spectral data were discarded from the HDX analysis. Finally, Deuteros 2.0 was used for statistical analysis of results exported from DynamX software.

Data availability

Co-crystal structure of compound **4** bound to human ROCK2. (2.63 Å, PDB ID: 9EP8, **https://www.rcsb.org/structure/unreleased/9EP8**) will be released after publication.

Conflicts of interest

There are no conflicts to declare.

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